Quadratic-Time Balanced Minimum Evolution Discrete Gradients, Column-Tiled SIMT Parallelism, and Topological Tree Optimization in Phylo2Vec Space

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Abstract

Balanced Minimum Evolution (BME) is one of the most statistically consistent and widely used distance-based criteria for phylogenetic tree inference, powering standard tools such as FastME. Classical BME optimizers represent trees as pointer-linked graph structures and perform sequential Subtree Prune and Regraft (SPR) hill-climbing, applying a single topological move per $O(N^2)$ sweep. In contrast, the Phylo2Vec representation bijectively encodes any rooted binary tree on N leaves as an integer vector $v \in \mathbb{Z}^{N-1}$ satisfying $0 \le v_i \le 2i$. In this paper, we present a complete mathematical formulation and high-performance software architecture for exact $O(N^2)$ BME discrete gradient computation and topological optimization directly in Phylo2Vec vector space. First, we derive the complete $(N-1) \times (2N-3)$ discrete gradient matrix $\nabla_{\text{disc}} L(v)$ over all valid coordinate values $0 \le j \le 2i$, paired with an $O(N)$ tree-based top-down cherry-first gauge transformation and an inverse tree-to-vector bijection. Second, we introduce a node-disjoint multi-coordinate (K-SPR) batch optimizer with dual-mode gauge rotation and stochastic 2-SPR basin hopping that applies multiple non-interfering SPR moves per $O(N^2)$ gradient evaluation. Third, to scale to large trees on multi-core CPUs and GPUs, we prove four architectural theorems: (i) the *Column-Independence Theorem* for lockstep SIMT clade-distance dynamic programming, (ii) the *Stackless Ancestor-Walk Top-Down Theorem* ($O(N \log N)$ expected work on balanced trees, $O(N^2)$ worst-case on caterpillars, with $O(1)$ stack space), (iii) the *Branchless Cherry-First Tree Decoding and Heavy-First* $\lceil \log_2(2N) \rceil$*-Slot Register-Stack Theorem* with 3-edge incident zeroing, $O(1)$ incremental destination mapping, and race-free SIMT parallel destination indexing, and (iv) the *Complete Column-Autonomy Theorem* for 4× memory-reduced column tiling paired with fused top-1 row candidate reductions. Across 120 benchmark datasets ($N = 20 \ldots 200$), our optimizer achieves a 99.2% win/tie rate (119/120 trials) in BME tree length against FastME while requiring up to 3.32× fewer $O(N^2)$ evaluations and running up to 4.16× faster on a single core. At large scale ($N = 100 \ldots 4{,}000$), our column-parallel and GPU-accelerated kernels achieve up to a 9.01× `subdist` speedup (5.98× end-to-end hybrid speedup) on an Apple M5 Metal GPU and provide an analytical multi-GPU scaling path up to $N = 350{,}000$–$400{,}000$ taxa on 8× NVIDIA H100 and 8× NVIDIA RTX PRO 6000 nodes.

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1 Introduction

Under the Balanced Minimum Evolution (BME) framework introduced by Pauplin [1] and formalized by Desper and Gascuel [2], the standard BME length $L(T)$ of an unrooted binary phylogenetic tree T with N leaves and symmetric pairwise leaf distance matrix $D \in \mathbb{R}^{N\times N}$ ($D_{xx} = 0$) is defined over unordered leaf pairs as:

$$L(T) = \frac{1}{2}\sum_{x=0}^{N-1}\sum_{\substack{y=0\\y\neq x}}^{N-1} D_{xy}\, 2^{-d_T(x,y)} = \sum_{0\le x<y\le N-1} D_{xy}\, 2^{-d_T(x,y)} = \sum_{x=0}^{N-1}\sum_{\substack{y=0\\y\neq x}}^{N-1} D_{xy}\, 2^{-d_{\text{edge}}(x,y)}, \quad (1)$$

where $d_T(x, y)$ denotes the number of internal degree-3 nodes on the unique path between leaf x and leaf y in T, and $d_{\text{edge}}(x, y) = d_T(x, y) + 1$ is the number of edges on that path. For convenience in discrete gradient recurrences, we also define the ordered-pair sum $L_{\text{ord}}(T) = 2L(T) = \sum_{x\neq y} D_{xy}2^{-d_T(x,y)}$; our standalone discrete gradient routines (`calculate_discrete_gradient_c` and `calculate_discrete_gradient_full_c`) report $\Delta L_{\text{ord}} = 2\Delta L$, whereas the topological optimizers (`optimize_phylo2vec_bme_c` and `optimize_fastme_bme_c`) report the standard unordered Pauplin length $L(T)$ in Eq. (1). Desper and Gascuel [2] and Lefort et al. [3] (FastME 2.0) showed that all $O(N^2)$ Subtree Prune and Regraft (SPR) moves on T can be evaluated in $O(N^2)$ total time via a two-pass recursive tree traversal that maintains balanced clade-to-clade average distances.

While FastME's $O(N^2)$ per-sweep complexity is asymptotically optimal for evaluating all SPR moves of a fixed tree, pointer-based graph representations impose three fundamental computational limitations:

1. **Single-Move Sequential Trajectory ($\Theta(N^3)$ Scaling):** FastME identifies the single best SPR move across all $O(N^2)$ candidates, mutates the pointer graph for that one move, and recomputes the entire $(2N-2)\times(2N-2)$ balanced averages table from scratch on the next iteration. Because optimizing an N-leaf tree requires $\Theta(N)$ SPR moves, total optimization time scales as $\Theta(N^3)$ sequential sweeps.

2. **Irregular Pointer Chasing:** Representing trees via heap-allocated node and edge structs prevents SIMD vectorization and SIMT GPU execution, causing severe CPU cache and TLB misses once the clade-distance table exceeds L2/L3 cache ($N \geq 1{,}500$).

3. **Lack of Algebraic Coordinate Gradients:** Pointer graphs do not expose a fixed-dimensional coordinate domain for discrete gradient descent, batch coordinate updates, or variational phylogenetic inference.

Recently, Penn et al. [4] introduced **Phylo2Vec**, a bijective representation that maps any rooted binary phylogenetic tree on N labeled leaves $\{0, 1, \ldots, N-1\}$ to an integer vector $v = (v_0, v_1, \ldots, v_{N-2}) \in \mathcal{V}$, where:

$$\mathcal{V} = \left\{(v_0, v_1, \ldots, v_{N-2}) \in \mathbb{Z}^{N-1} \;\middle|\; 0 \le v_i \le 2i \quad \forall\, i \in \{0, \ldots, N-2\}\right\}, \qquad |\mathcal{V}| = (2N-3)!!. \quad (2)$$

Perturbing a single coordinate $v_i \leftarrow j$ ($j \in \{0, \ldots, 2i\}$) corresponds to a structured SPR move in which the subtree attached at step i is pruned and regrafted onto the edge indexed by j in the prefix tree on leaves $\{0, \ldots, i\}$.

1.1 Contributions and Architecture Overview

In this paper, we describe a complete, self-contained mathematical and software framework for BME discrete gradient computation and topological tree optimization in Phylo2Vec space:

1. **Full $0 \le j \le 2i$ Discrete Gradient in $O(N^2)$ Time (Section 2):** For any tree in cherry-first canonical gauge ($0 \le v_i \le i$), we compute the complete $(N-1) \times (2N-3)$ discrete gradient matrix $\Delta_{i,j} = L_{\text{ord}}(v^{(i \leftarrow j)}) - L_{\text{ord}}(v)$ across all $2i+1$ valid regrafting coordinates $0 \le j \le 2i$—covering leaf-incident pendant edges ($0 \le j \le i$), internal clade edges ($i+1 \le j \le 2i-2$), and root-incident edges ($j \in \{2i-1, 2i\}$ with the boundary guard $2i-1 > i$ at row $i = 1$).

2. **Cherry-First Gauge Transformation and Exact Inverse Bijection (Section 2.2):** We prove that every decoded binary tree can be relabeled into the cherry-first submanifold $\mathcal{V}_{\text{cf}} = \{v \in \mathcal{V} : 0 \le v_i \le i\}$ in $O(N)$ time via a single top-down pass that issues leaf labels in birth order (`relabel_from_rooted_tree`), and we provide an exact inverse bijection (`encode_rooted_to_v_c` and `encode_unrooted_to_v_c`) that recovers $v \in \mathcal{V}$ from any rooted or unrooted tree topology.

3. **Node-Disjoint Multi-Coordinate (K-SPR) Batch Optimizer (Section 3):** We design a hybrid optimizer that extracts multiple mutually node-disjoint improving coordinate moves from a single discrete gradient evaluation, applying K non-interfering SPR moves simultaneously with exact line-search verification, dual-mode gauge rotation, and 2-SPR basin hopping.

4. **SIMT GPU, Multi-Core CPU, and Column-Tiled Algorithms (Section 4):** We prove four theorems that transform the tree-based recurrences into lockstep vector/matrix operations: the *Column-Independence Theorem*, the *Stackless Ancestor-Walk Theorem*, the *Heavy-First* $\lceil \log_2(2N) \rceil$*-Slot Register-Stack Theorem* with $O(1)$ sequential and race-free SIMT destination mapping, and the *Complete Column-Autonomy Theorem* for $4\times$ memory-reduced column tiling.

2 Mathematical Formulation of the Phylo2Vec Discrete Gradient

2.1 Phylo2Vec Tree Decoding and Unrooted Representation

Given a vector $v \in \mathcal{V}_{\text{cf}}$ in the cherry-first canonical domain ($0 \le v_i \le i$ for all $i \in \{0, \dots, N-2\}$), the corresponding rooted binary tree T_{root} with $2N-1$ nodes (leaves $0 \dots N-1$ and internal nodes $N \dots 2N-2$) is decoded in $O(N)$ time in reverse coordinate order $k = 0, \dots, N-2$ (where $i = N-2-k$):

$$\text{pairs}[k] = (p_1, p_2) = (v_i,\ i+1), \qquad m = N + k. \tag{3}$$

Using a path-compressed union-find array parents_uf$[0 \dots 2N-2]$ initialized to the identity, each step k finds the current subtree roots $r_1 = \text{find}(p_1)$ and $r_2 = \text{find}(p_2)$, creates internal node $m = N + k$ with children $c_1(m) = r_2$ and $c_2(m) = r_1$, and sets parents_uf$[r_1]$ = parents_uf$[r_2] = m$. (For general vectors $v \in \mathcal{V}$ with $v_i > i$, forward array-insertion decoding constructs T_{root} in $O(N^2)$ worst-case time.)

Because the BME tree length $L(T)$ in Eq. (1) is invariant to the placement of the root on the top edge, we collapse the rooted root node $2N-2$ (whose children are $R = 2N-3$ and $c_{\text{other}} = \min(c_1(2N-2), c_2(2N-2))$) into an unrooted tree T on $M = 2N-2$ nodes $\{0, 1, \dots, 2N-3\}$ rooted at the degree-3 trifurcating node $R = 2N-3$ with three children:

$$R_{c_0} = c_1(R), \qquad R_{c_1} = c_2(R), \qquad R_{c_2} = c_{\text{other}}. \tag{4}$$

Every internal node $u \in \{N, \dots, R-1\}$ has two downward children $c_1(u), c_2(u)$ and one upward parent parent_u$[u]$, forming a 3-regular adjacency table adj_u$[u, 0..2] = (c_1(u), c_2(u), \text{parent_u}[u])$, while adj_u$[R, 0..2] = (R_{c_0}, R_{c_1}, R_{c_2})$. We compute subtree node counts sub_size$[u]$ bottom-up

and assign contiguous Depth-First Search (DFS) entry times $\mathrm{tin}[u] \in \{0, \ldots, 2N-3\}$ top-down:

$$\mathrm{tin}[R] = 0, \quad \mathrm{tin}[R_{c_0}] = 1, \quad \mathrm{tin}[R_{c_1}] = 1 + \mathrm{sub_size}[R_{c_0}], \quad \mathrm{tin}[R_{c_2}] = \mathrm{tin}[R_{c_1}] + \mathrm{sub_size}[R_{c_1}],$$
$$\mathrm{tin}[c_1(u)] = \mathrm{tin}[u] + 1, \quad \mathrm{tin}[c_2(u)] = \mathrm{tin}[u] + 1 + \mathrm{sub_size}[c_1(u)] \qquad (u = R-1 \text{ down to } N). \tag{5}$$

By construction, a node v lies in the strict downward subtree $T_u \setminus \{u\}$ of internal node u if and only if $\mathrm{tin}[u] < \mathrm{tin}[v] < \mathrm{tin}[u] + \mathrm{sub_size}[u]$, and for $v \in T_u \setminus \{u\}$, v lies in $T_{c_1(u)}$ if and only if $\mathrm{tin}[v] < \mathrm{tin}[c_2(u)]$.

2.2 Cherry-First Gauge Transformation and Exact Inverse Bijection

While general Phylo2Vec vectors satisfy $0 \le v_i \le 2i$, the $O(N)$ bottom-up union-find decoding above requires the *cherry-first canonical gauge* $0 \le v_i \le i$ (in which every merge occurs on a leaf pendant edge rather than an internal ancestral edge). We establish that any already-decoded rooted binary tree can be transformed into cherry-first gauge in $O(N)$ time via a leaf permutation $\pi \in S_N$ (`relabel_from_rooted_tree`), and conversely any rooted or unrooted tree can be encoded back into $v \in \mathcal{V}$ ($0 \le v_i \le 2i$).

Theorem 2.1 (Tree-Based $O(N)$ Cherry-First Gauge Transformation and Clade-Flip Duality)**.** *Let T be any rooted binary tree on N leaves with child arrays c_1, c_2 and root $2N-2$, whose internal nodes $u \in \{N, \ldots, 2N-2\}$ are indexed so that every parent has a larger index than its children (as produced by Phylo2Vec decoding). For every node $u \in \{0, \ldots, 2N-2\}$, define the minimum leaf in u's clade bottom-up by $m(x) = x$ for $x < N$ and $m(u) = \min(m(c_1(u)), m(c_2(u)))$ for $u \ge N$. Fix a clade orientation flip_mode $\in \{0, 1\}$, set $\ell(2N-2) = 0$ and a label counter $t = 1$, and visit the internal nodes top-down in decreasing index order $u = 2N-2, 2N-3, \ldots, N$:*

- *If flip_mode $= 0$, order the children (a, b) of u so that $m(a) < m(b)$; if flip_mode $= 1$, order (a, b) so that $m(a) > m(b)$.*
- *Set $\ell(a) = \ell(u)$ and $\ell(b) = t$, record the cherry-first coordinate $v^{cf}_{t-1} = \ell(a)$, and increment t.*

Then $\pi(k) = \ell^{-1}(k)$ defines a bijection on $\{0, \ldots, N-1\}$ computed in $O(N)$ time from T such that $v^{cf} \in \mathcal{V}_{cf}$ ($0 \le v^{cf}_i \le i$ for all i) encodes the exact same tree T under the permuted distance matrix $D_\pi[a, b] = D[\pi(a), \pi(b)]$, with $L(v^{cf}; D_\pi) = L(v; D)$. If T was decoded from some $v \in \mathcal{V}_{cf}$, then flip_mode $= 0$ returns $v^{cf} = v$ and $\pi = id$. (When starting from an undecoded general vector $v \in \mathcal{V}$ with $v_i > i$, initial array-insertion decoding into T and explicit materialization of the $N \times N$ matrix D_π take $O(N^2)$ time.)

Proof. Write $t(u) = \ell(b)$ for the label issued at internal node u. The counter issues each of $1, \ldots, N-1$ exactly once. Every label, including the root label 0, is passed unchanged from a node to its a-child and so travels down a chain of a-children to exactly one leaf; since each node receives exactly one label from its parent, ℓ restricted to the leaves is a bijection onto $\{0, \ldots, N-1\}$. The label $\ell(a) = \ell(u)$ was issued at a proper ancestor of u (or is the root label 0), hence before $t(u)$, so $v^{\mathrm{cf}}_{t(u)-1} = \ell(u) < t(u)$ and $v^{\mathrm{cf}} \in \mathcal{V}_{\mathrm{cf}}$. Because parents are visited before their children, the issue order is heap-ordered: $t(w) > t(u)$ for every internal descendant w of u.

Decoding v^{cf} processes $i = N-2$ down to 0, merging the clades that contain leaves v^{cf}_i and $i+1$. We show by induction that step i creates exactly the clade T_u of the node with $t(u) = i+1$. Every internal node of T_a and T_b is a descendant of u, so by heap order it has $t > i+1$ and its merge was completed at an earlier step, whereas u itself has not yet been processed; thus T_a and T_b are both complete, separate components. Leaf $\ell(b) = i+1$ lies in T_b and leaf $\ell(a) = v^{\mathrm{cf}}_i$ lies in T_a, so step i merges precisely T_a and T_b, producing T_u. After step 0 every clade of T has been produced, so v^{cf} encodes T under the labels ℓ, i.e., under D_π, and the BME length is unchanged. The argument never uses which child is a, so it holds for both values of flip_mode. Finally, if

T was decoded from $v \in \mathcal{V}_{\text{cf}}$, the node created at step i has index $N + (N - 2 - i)$ and children with minimum leaves v_i and $i + 1$ (all other leaves in these two components exceed $i + 1$); it is therefore the $(i + 1)$-th node visited, flip_mode $= 0$ issues $t = i + 1$ to the child containing leaf $i + 1$, and induction on i gives $\ell = \text{id}$ and $v^{\text{cf}} = v$. □

Intuitively, v^{cf} is a birth history of T: reading the coordinates forward, leaf $i + 1$ is born at time $i + 1$ from the existing lineage v_i^{cf}, the internal-node index order supplies the ranking of birth events, and at every split the a-child continues its parent's lineage label while the b-child receives the next label. The labels must follow a top-down ranking: assigning T_b a contiguous depth-first interval starting at $\ell(u) + |T_a|$ violates heap order whenever T_a contains an internal node (e.g., for $((x_0, x_1), (x_2, x_3))$ it yields $(0, 0, 2)$, which decodes to a caterpillar).

Theorem 2.2 (Exact Inverse Bijection from Rooted and Unrooted Trees to $\mathcal{V}$)**.** *Given any rooted binary tree T on leaves $\{0, \ldots, N-1\}$ (or any unrooted binary tree T canonically rooted on the pendant edge $(0, adj_u[0, 0])$ incident to leaf 0 via post-order traversal in* `encode_unrooted_to_v_c`*), let each internal node $m \in \{N, \ldots, 2N-2\}$ combine child clades $c_1(m), c_2(m)$ with minimum leaves $l_{\min}(m) = \min(m(c_1(m)), m(c_2(m)))$ and $l_{\max}(m) = \max(m(c_1(m)), m(c_2(m)))$. Then $l_{\max}(m)$ takes each value in $\{1, \ldots, N-1\}$ over the $N-1$ internal nodes. Processing step $i = 0, \ldots, N-2$ (for leaf $i+1$) along the ancestor chain from leaf $i+1$'s parent $p = parent[i+1]$ up to the current active root in the prefix tree on $\{0, \ldots, i\}$ (*`encode_rooted_to_v_c`*):*

- *If p is the first active ancestor encountered on the lineage of sibling minimum leaf $s = m(sib(i + 1)) < i + 1$, then leaf $i + 1$ attaches directly to the pendant edge of leaf s, so $v_i = s \in \{0, \ldots, i\}$.*
- *Otherwise, walking down the active ancestor chain of leaf s to locate the predecessor internal node q whose parent in the prefix tree is p yields the internal edge label $v_i = edge_label[q] \in \{i + 1, \ldots, 2i\}$, after which p is activated with $edge_label[p] = 2i + 1$.*

This recovers the unique vector $v \in \mathcal{V}$ in $O(N)$ time on cherry-first trees and $O(N \cdot height(T))$ time ($O(N \log N)$ expected, $O(N^2)$ worst-case) on general rooted trees.

2.3 Two-Pass $O(N^2)$ Balanced Clade Distance Matrix (`subdist`)

To evaluate all SPR moves in $O(N^2)$ time, we construct the symmetric $(2N - 2) \times (2N - 2)$ balanced clade distance matrix subdist$[u, v]$, which stores the directional balanced average distance between the subtrees induced by every pair of nodes (u, v) in the unrooted tree T:

1. **Leaf Initialization ($0 \le x, y < N$):** subdist$[x, y] = D_{xy}$.
2. **Bottom-Up Clade Recurrence ($u = N, \ldots, R - 1$):** For any node $v \notin T_u$, the balanced average distance from u's downward subtree T_u to v satisfies:
$$\text{subdist}[u, v] = \text{subdist}[v, u] = \frac{1}{2} \left(\text{subdist}[c_1(u), v] + \text{subdist}[c_2(u), v]\right). \tag{6}$$
3. **Trifurcating Root 3-Way Recurrence ($u = R = 2N - 3$):** For each child R_{c_k} ($k \in \{0, 1, 2\}$) with complementary root children $R_{c_{k'}}, R_{c_{k''}}$ ($k', k'' \neq k$), every node $v \in T_{R_{c_k}}$ has:
$$\text{subdist}[R, v] = \text{subdist}[v, R] = \frac{1}{2} \left(\text{subdist}[R_{c_{k'}}, v] + \text{subdist}[R_{c_{k''}}, v]\right). \tag{7}$$
4. **Top-Down Complementary Clade Recurrence ($p = R - 1$ down to N):** When $v \in T_p \setminus \{p\}$ lies inside the downward subtree of p, the clade at p oriented *away* from v consists of p's parent $g = \text{parent_u}[p]$ and the sibling child $\text{sib}_p(v) \in \{c_1(p), c_2(p)\}$ not containing v:
$$\text{subdist}[p, v] = \text{subdist}[v, p] = \frac{1}{2} \left(\text{subdist}[\text{sib}_p(v), v] + \text{subdist}[\text{parent_u}[p], v]\right). \tag{8}$$

5. **Diagonal and Local Neighbor Distances (`nd_u`):** Setting $\mathrm{subdist}[u,u]=0$ for all u, we cache for each internal node $u \in \{N,\dots,R\}$ with neighbors $(n_0,n_1,n_2)=\mathrm{adj_u}[u,0..2]$ the three pairwise neighbor-clade distances:

$$\mathrm{nd_u}[u,0]=\mathrm{subdist}[n_1,n_2],\quad \mathrm{nd_u}[u,1]=\mathrm{subdist}[n_0,n_2],\quad \mathrm{nd_u}[u,2]=\mathrm{subdist}[n_0,n_1]. \tag{9}$$

2.4 Half-Tree SPR Recurrence and the Full $0 \le j \le 2i$ Gradient Mapping

For each row $i \in \{1,\dots,N-2\}$ (corresponding to leaf index $v_{\mathrm{index}}=i+1$), the subtree pruned when coordinate v_i changes is $T_{\mathrm{prune}}=\mathrm{v_map}[v_{\mathrm{index}}]$, which was attached to internal node $P=\mathrm{parent_u}[T_{\mathrm{prune}}]$ (or c_{other} if $T_{\mathrm{prune}}=R$). Removing T_{prune} and contracting P leaves an edge between P's other two neighbors, other_0 and other_1, partitioning the remaining tree into two disjoint half-trees rooted at other_0 and other_1.

Traversing into the half-tree starting at $u_0=\mathrm{other}_0$ (with opposite anchor $\mathrm{orig}=\mathrm{other}_1$ and initial distance $d_{kC}^{(0)}=\mathrm{subdist}[T_{\mathrm{prune}},\mathrm{orig}]$), each step across an internal node u from predecessor C branches into the two outgoing neighbors b_0,b_1 of u with path-length weight $w_\ell=2^{-(\ell+1)}$. Updating the path-adjusted cross-clade distance D_{AC} and pruned-clade distance d_{kC}:

$$\begin{aligned}
D_{AC}(b_0) &= \mathrm{subdist}[C,b_1]^{\mathrm{before}} + (\mathrm{subdist}[\mathrm{orig},b_1]-\mathrm{subdist}[T_{\mathrm{prune}},b_1])\,2^{-(\ell+1)},\\
\Delta L_{\mathrm{ord}}(u\to b_0) &= \Delta L_{\mathrm{ord}}(C\to u)+\frac{1}{4}\left(D_{AC}(b_0)+\mathrm{subdist}[T_{\mathrm{prune}},b_0]-D_{AB}(u)-d_{kC}(u)\right),\\
d_{kC}(b_0) &= \frac{1}{2}\left(d_{kC}(u)+\mathrm{subdist}[T_{\mathrm{prune}},b_1]\right),
\end{aligned} \tag{10}$$

and symmetrically for b_1, we store $\Delta L_{\mathrm{ord}}(u\to b)$ in $\mathrm{edge_delta}[c_{\mathrm{end}}(u,b)]$, where $c_{\mathrm{end}}(u,b)\in\{0,\dots,2N-3\}$ is the child endpoint of the edge (u,b) in the unrooted tree rooted at R. Running this recurrence on both $(\mathrm{other}_0,\mathrm{other}_1)$ and $(\mathrm{other}_1,\mathrm{other}_0)$ populates $\mathrm{edge_delta}[0\dots 2N-3]$ in $O(N)$ time per row i.

Theorem 2.3 (Complete $0 \le j \le 2i$ Edge-to-Coordinate Mapping)**.** *For any cherry-first vector $v \in \mathcal{V}_{cf}$ ($v_i \le i$) and row $i \in \{1,\dots,N-2\}$, the change in ordered BME tree length $\Delta_{i,j}=L_{ord}(v^{(i\leftarrow j)})-L_{ord}(v)=2(L(v^{(i\leftarrow j)})-L(v))$ for all $2i+1$ valid Phylo2Vec coordinates $j\in\{0,\dots,2i\}$ is given by:*

$$\Delta_{i,j}=\begin{cases} edge_delta[dest_row_i[j]], & 0\le j\le i,\\ edge_delta[(2N-2)-2i+j], & i+1\le j\le 2i-2,\\ edge_delta[c_{other}], & j\in\{2i-1,\,2i\}\cap[i+1,\,2i], \end{cases} \tag{11}$$

where $dest_row_i[j]$ is the highest ancestor of leaf j formed strictly after step i in the bottom-up decoding order ($m\in\{N,\dots,2N-3-i\}$), and $c_{other}=\min(c_1(2N-2),c_2(2N-2))$ is the non-R child of the rooted root $2N-2$ representing the collapsed top edge of the unrooted tree. In particular, at row $i=1$, $2i-1=1=i$ belongs to the leaf-pendant range $[0,i]$, so $j=1$ receives $edge_delta[dest_row_1[1]]$ (guarded by `if (2 * i - 1 > i)`) while $j=2i=2$ receives $edge_delta[c_{other}]$.

Proof. At step i of the Phylo2Vec decoding process, $N-2-i$ internal merges have already occurred (corresponding to internal nodes $m\in\{N,\dots,2N-3-i\}$). For $0\le j\le i$, regrafting onto coordinate j attaches T_{prune} to the current root of the clade containing leaf j, which is $\mathrm{dest_row}_i[j]$. For $i+1\le j\le 2i$, Phylo2Vec indexes internal clades in reverse order of their creation time: $m(i,j)=N+(2i-j)$. Because the unrooted tree collapses the top edge $(2N-2)$ into (R,c_{other}) and shifts internal node indices above the pruned merge by $N-2-i$, the $i-2$ non-root internal edges map bijectively to $u=(2N-2)-2i+j$ for $i+1\le j\le 2i-2$, while the rooted edges incident to the root of the prefix tree that satisfy $j>i$ ($j\in\{2i-1,2i\}$ for $i\ge 2$, and $j=2$ for $i=1$) correspond to the unrooted edge indexed by c_{other}. □

3 Topological Optimization via Node-Disjoint K-SPR Search

Evaluating the full $(N-1) \times (2N-3)$ discrete gradient matrix $\Delta = \nabla_{\text{disc}} L(v)$ reveals the exact BME change of all $(N-1)^2$ single-coordinate SPR moves in a single $O(N^2)$ sweep. Rather than applying only a single coordinate update per sweep, we exploit the discrete gradient matrix to apply $K \geq 1$ non-interfering SPR moves simultaneously.

3.1 Node-Disjoint Path Corridors and Batch Line Search

Each candidate coordinate update $v_i \leftarrow j^*$ ($j^* = \arg\min_{0 \leq j \leq 2i} \Delta_{i,j}$ with $\Delta_{i,j^*} < -\epsilon$) corresponds to moving the pruned subtree $T_{\text{prune}}(i) = \text{v_map}[i+1]$ from its current attachment node $P(i) = \text{parent_u}[T_{\text{prune}}(i)]$ to target edge endpoint $U(i, j^*)$ along a unique path corridor $\mathcal{P}_T(P(i), U(i, j^*))$ in the unrooted tree T.

In the BME length formula (Eq. (1)), an SPR move along $\mathcal{P}_T(P(i), U(i, j^*))$ alters the topological path weights $2^{-d_T(x,y)}$ primarily for paths passing through $\mathcal{P}_T(P(i), U(i, j^*))$. Consequently, if a set of improving coordinate moves $\mathcal{M} = \{(i_1, j_1^*), \ldots, (i_K, j_K^*)\}$ satisfies:

1. **Vertex-Disjointness of Corridors:** $\mathcal{P}_T(P(i_a), U(i_a, j_a^*)) \cap \mathcal{P}_T(P(i_b), U(i_b, j_b^*)) = \emptyset$ for all $a \neq b$ (including source and target neighbors), and

2. **Coordinate Prefix Independence:** No move i_a modifies a subtree minimum-leaf index referenced by another move i_b,

then the K moves are topologically non-interfering and their combined BME improvement satisfies $\Delta L(\mathcal{M}) \approx \frac{1}{2} \sum_{k=1}^{K} \Delta_{i_k, j_k^*} < 0$.

Our greedy batch selector (`optimize_phylo2vec_bme_c`) sorts all improving rows i in ascending order of Δ_{i,j^*} (most negative first), marks the tree nodes on $\mathcal{P}_T(P(i), U(i, j^*))$ in a boolean mask blocked$[0 \ldots 2N-3]$ as each move is accepted into $\mathcal{M}$, and then performs a fast backtracking line search over $K' \in \{|\mathcal{M}|, \lfloor |\mathcal{M}|/2 \rfloor, \ldots, 1\}$ using $O(N)$ cherry-first gauge canonicalization and exact $O(N^2)$ BME evaluation, guaranteeing strict monotonic descent $L(v^{(t+1)}) < L(v^{(t)})$ on every step and exact single-move fallback when $K' = 1$.

3.2 Complete 4-Stage Phylo2Vec BME Optimization Pipeline

The complete topological optimizer (`optimize_phylo2vec_bme_c`) combines four stages:

1. **Stage A (Full $0 \leq j \leq 2i$ Discrete Gradient + Node-Disjoint K-SPR Batch Descent):** Computes $\nabla_{\text{disc}} L(v)$ via Theorem 2.3 and applies node-disjoint multi-coordinate updates until $\min_{i,j} \Delta_{i,j} \geq -10^{-11}$. Whenever a selected coordinate has $j^* > i$, Theorem 2.1 re-canonicalizes $v^{(i \leftarrow j^*)}$ back to the cherry-first domain $v_i^{\text{cf}} \leq i$.

2. **Stage B (Dual-Mode $O(N)$ Cherry-First Gauge Rotation):** When a local minimum is reached in the current leaf labeling, the optimizer applies Theorem 2.1 with alternating clade-orientation parity (flip_mode $= 1$ vs. flip_mode $= 0$). Reversing the left-right clade assignment swaps which child subtree acts as the movable leaf $i+1$ versus the anchor clade at every internal node, exposing a complementary set of Phylo2Vec coordinate moves without changing the tree topology.

3. **Stage C (Remaining Unrooted SPR Scan + Exact Inverse Bijection):** Single-coordinate moves in the two gauges of Stage B reach only part of the unrooted SPR neighborhood. Enumerating all $2(N-3)(2N-7)$ distinct SPR neighbors by brute force (`verify_spr_coverage.py`, 30 random trees per N), each gauge reaches 27–33% of them and the two gauges together 52–56% for $N = 8 \ldots 20$, and no tested tree was fully covered. Stage C therefore scans the remaining moves directly on the unrooted tree: it prunes every clade that is not indexed by a coordinate v_i ($i \geq 1$), together with the complementary

clade above every internal node, and evaluates all of their regraft edges with the same `subdist` table and half-tree recurrence (Eq. (10)) in $O(N^2)$ time. The best improving move is applied, the tree is encoded back into $v \in \mathcal{V}$ via Theorem 2.2 (`encode_unrooted_to_v_c`), and Stage A resumes.

4. **Stage D (Stochastic 2-SPR Basin-Hopping Kicks):** On noisy distance matrices ($\sigma > 0$), single-SPR local minima can trap greedy hill-climbers. After convergence, the optimizer applies a random 2-SPR coordinate perturbation and repeats the descent from the perturbed state, retaining the best topology found.

4 Parallel Multi-Core CPU, SIMT GPU, and Column-Tiled Architecture

At large N ($N = 1{,}000 \ldots 400{,}000$), naive parallelization of tree dynamic programming fails due to strided column scatter writes, per-level synchronization barriers, and $O(N)$ per-thread BFS queue memory. In `bench_parallel_gpu.m` and `phylo2vec_cuda_8xh100.cu`, we prove and implement four theorems that reformulate the Phylo2Vec discrete gradient into lockstep, cache-resident vector and matrix operations.

4.1 Column-Independence Theorem for Lockstep SIMT Warps

Theorem 4.1 (Column-Independence of `subdist`)**.** *In the bottom-up recurrence* $subdist[u, x] = \frac{1}{2}(subdist[c_1(u), x] + subdist[c_2(u), x])$ *($u = N, \ldots, R-1$), every entry in column $x \in \{0, \ldots, R-1\}$ depends exclusively on prior rows $c_1(u), c_2(u) < u$ within the* ***exact same column*** *x.*

Theorem 4.1 has two major architectural consequences for SIMT GPUs (Apple Metal and NVIDIA CUDA) and multi-core CPUs:

1. **Zero Per-Level Synchronization Barriers and** 100% **Coalesced SIMT Warps:** Instead of parallelizing across rows u at the same tree depth (which requires launching a separate GPU kernel at every level of the tree and reading non-contiguous columns), we assign thread gid $= x$ to column x. All 32 threads in a GPU warp iterate over $u = N, \ldots, R-1$ in lockstep (`k_bottom_up_leaf_cols` and `k_bottom_up_int_cols`), issuing 100% coalesced 128-byte cache-line reads and writes (`subdist[u * n_nodes_u + x]`) with zero branch divergence.

2. **Elimination of Strided Scatter Writes via Shared-Memory Tiled Transpose:** Neither the bottom-up pass nor the top-down pass performs strided column scatter writes (`subdist[v * n_nodes_u + u]`) inside the inner loop; instead, a 16×16 shared-memory tiled transpose (`k_transpose_leaf_block` / `gpu_transpose_leaf_block`) copies the rectangular leaf block $[N \ldots R-1,\ 0 \ldots N-1]$ into $[0 \ldots N-1,\ N \ldots R-1]$ between Pass 1 and Pass 2 with fully coalesced global memory reads and writes.

4.2 Stackless Ancestor-Walk Top-Down Theorem

Theorem 4.2 (Stackless Ancestor-Walk Top-Down Pass)**.** *In the top-down recurrence (Eq. (8)), entry $subdist[p, v]$ is updated if and only if $v \in T_p \setminus \{p\}$, i.e., if and only if p is a strict ancestor of v in the unrooted tree rooted at R. Moreover, given the Euler-tour entry times tin in Eq. (5), thread $v \in \{0, \ldots, R-1\}$ can traverse the exact top-down ancestor path from R to v in $depth(v)$ steps with $O(1)$* ***register memory and zero stack space****:*

$$next_curr = \begin{cases} c_1(p), & \text{if } tin[v] < tin[c_2(p)], \\ c_2(p), & \text{otherwise}, \end{cases} \qquad sib = \begin{cases} c_2(p), & \text{if } tin[v] < tin[c_2(p)], \\ c_1(p), & \text{otherwise}, \end{cases} \tag{12}$$

writing both subdist$[p, v]$ and the symmetric entry subdist$[v, p]$ in-place without data races across threads v. Across all $2N-2$ threads, total work is $\sum_{v=0}^{2N-3} depth(v)$, which is $O(N \log N)$ in expectation on balanced or Yule–Harding random trees ($\max_v depth(v) = O(\log N)$) and $O(N^2)$ in the worst case on degenerate caterpillar trees ($\max_v depth(v) = O(N)$).

Proof. Because $T_{c_1(p)}$ occupies $[\text{tin}[p] + 1, \text{tin}[c_2(p)))$ and $T_{c_2(p)}$ occupies $[\text{tin}[c_2(p)], \text{tin}[p] + \text{sub_size}[p])$, the single comparison $\text{tin}[v] < \text{tin}[c_2(p)]$ determines which child of p contains v. Furthermore, at ancestor p of v, thread v reads subdist$[\text{sib}, v]$ (where $T_{\text{sib}} \cap T_p^{\text{child}} = \emptyset$, so subdist$[\text{sib}, v]$ was already finalized in the bottom-up pass) and subdist$[\text{parent_u}[p], v]$ (computed by thread v on the immediately preceding step of its top-down walk). Conversely, the symmetric write subdist$[v, p]$ modifies column p at row $v \in T_p \setminus \{p\}$, which is never read by thread p because thread p only reads rows outside T_p. Thus all threads $v \in \{0, \ldots, R-1\}$ execute concurrently with zero race conditions. □

4.3 Heavy-First $\lceil \log_2(2N) \rceil$-Slot Register Stack and Race-Free Destination Updates

Theorem 4.3 (Branchless Cherry-First Tree Decoding, Logarithmic Stack Bound, and Race-Free Destination Mapping)**.** *The tree setup and Stage 3 SPR sweep satisfy the following three structural invariants:*

1. ***Branchless*** $O(N)$ ***Cherry-First Component-Root Invariant:*** *For any cherry-first vector $v \in \mathcal{V}_N^{cf}$ ($0 \le v_i \le i$), at merge step $k = 0, \ldots, N-2$ ($i = N-2-k$), the active forest partitions $\{0, \ldots, N-1\}$ into $i+2$ connected components $C_0, \ldots, C_{i+1}$ whose minimum leaves are strictly $\min(C_x) = x$. Consequently, initializing root$[x] \leftarrow x$ for $x \in \{0, \ldots, N-1\}$ allows the rooted tree topology $(c_1(m), c_2(m))$, minimum-leaf table $l_{\min}(m) = v_i$, and pruned-subtree lookup v_map$[i+1]$ = root$[i+1]$ to be constructed in a single branchless $O(N)$ loop (child$_1[N+k] \leftarrow$ root$[i+1]$, child$_2[N+k] \leftarrow$ root$[v_i]$, root$[v_i] \leftarrow N+k$), completely eliminating the intermediate* `pairs` *array and Union-Find path compression.*

2. ***Logarithmic Stack Bound and 3-Edge Incident Zeroing:*** *In the Stage 3 half-tree traversal (*`run_half_tree_dfs`*), each internal node u visited from C branches into two neighbors b_0, b_1 with remaining component sizes $S(u \to b_k) =$ sub_size$[b_k]$ if parent_u$[b_k] = u$ and $(2N-2) -$ sub_size$[u]$ otherwise. Pushing the neighbor with the larger component size onto a LIFO stack first and the smaller neighbor second guarantees that the stack depth never exceeds $\lfloor \log_2(2N-2) \rfloor + 2$ for* ***any*** *binary tree topology (at most 16 slots for $N \le 32{,}768$ and at most 20 slots for $N \le 500{,}000$). Moreover, because every coordinate $0 \le j \le 2i$ maps to an edge outside $T_{\text{tree_root}}$ and* `run_half_tree_dfs` *overwrites every non-incident edge outside $T_{\text{tree_root}}$, only the three edges incident to parent (with child endpoints $e(w) = w$ if parent_u$[w]$ = parent else parent, for $w \in \{$tree_root, other$_0$, other$_1\}$) need to be zero-initialized prior to traversal, eliminating $O(N^2)$ redundant memory stores across all $N-2$ rows and preventing inter-warp write conflicts with other$_0$, other$_1$ half-tree traversals.*

3. *$O(1)$* ***Sequential and Race-Free Parallel Destination Mapping:*** *In bottom-up Phylo2Vec merge order $m = N + k$ ($k = 0, \ldots, N-3$), the larger minimum leaf of merge m is deterministically $l_{\max}(N+k) = N-1-k$. Consequently, as row index $i = v_{index} - 1$ steps downward from $N-2$ to 1 on a CPU, the destination lookup array dest_row$[0 \ldots N-1]$ (initialized to dest_row$[j] = j$ at $i = N-2$) changes by* ***only a single entry per row****:*

$$dest_row[l_{\min}(N+k)] \leftarrow N+k \qquad \text{for } k = N-2-i, \tag{13}$$

eliminating the $O(N^2)$ destinations matrix completely. For independent SIMT GPU thread-blocks executing row i with $n_{active} = N-2-i$ active merges, precomputing on the host the successor index next_k_same_lmin$[k] = \min\{k' > k : l_{\min}(N+k') = l_{\min}(N+k)\}$ (or N if none

exists) ensures that for each leaf column j, at most one active merge $k \in \{0, \dots, n_{active} - 1\}$ satisfies next_k_same_lmin$[k] \geq n_{active}$, allowing all threads k in a threadblock to populate $\Delta_{i,l_{\min}(N+k)} = $ edge_delta$[N + k]$ concurrently with ***zero write-after-write race conditions****.*

4.4 Complete Column-Autonomy Theorem for Low-Memory Column Tiling

Theorem 4.4 (Complete Column-Autonomy and 4× `subdist` Memory Reduction via Column Tiling)**.** *Suppose the bottom-up recurrence (Eq. (6)) and the stackless ancestor-walk top-down pass (Theorem 4.2) are executed on the N leaf columns $x \in \{0, \dots, N-1\}$ in tiles of width B_{tile} (e.g., $B_{tile} = 128$), storing only the internal-to-leaf table subdist_int_leaf$[w - N, x] = subdist[w, x]$ of dimension $(N-1) \times N$ ($\frac{1}{4}$ the size of subdist). Then for* ***any*** *node $w \in \{0, \dots, 2N-3\}$, its entire row row$_w[0 \dots 2N-3] = subdist[w, \cdot]$ can be reconstructed autonomously in $O(N + depth(w)) = O(N)$ time inside a single $2N$-element L1/shared-memory buffer (`compute_autonomous_row_f64` / `dev_compute_autonomous_row`):*

1. *Copy row$_w[0 \dots N-1] \leftarrow D[w, 0 \dots N-1]$ (if $w < N$) or subdist_int_leaf$[w-N, 0 \dots N-1]$ (if $w \geq N$).*
2. *For $u = N, \dots, R-1$: set row$_w[u] = \frac{1}{2}($row$_w[c_1(u)] +$ row$_w[c_2(u)])$.*
3. *Walk top-down along the depth(w) ancestors p from R to w, setting row$_w[p] = \frac{1}{2}($row$_w[sib_p(w)] +$ row$_w[$parent_u$[p]])$, and finally set row$_w[w] = 0$.*

Proof. By symmetry of subdist, $\text{row}_w[x] = \text{subdist}[w, x] = \text{subdist}[x, w]$ for all leaves $x \in \{0, \dots, N-1\}$. For any internal node $u \in \{N, \dots, R-1\}$: (i) if $T_u \cap T_w = \emptyset$, the bottom-up step averages subdist$[w, x]$ over $x \in T_u$, yielding subdist$[w, u]$; (ii) if $u \in T_w \setminus \{w\}$ is a strict descendant of w, then T_u lies entirely within a single child subtree of w, where every leaf $x \in T_u$ already holds its exact top-down distance $\text{subdist}[w, x] = \frac{1}{2}(\text{subdist}[\text{sib}_w(u), x] + \text{subdist}[\text{parent_u}[w], x])$; by linearity of balanced subtree averaging over $x \in T_u$, the bottom-up pass on row_w automatically yields the exact top-down value $\text{row}_w[u] = \text{subdist}[w, u]$ for all $u \in T_w \setminus \{w\}$; and (iii) if u is a strict ancestor of w ($w \in T_u \setminus \{u\}$), Step 3 overwrites $\text{row}_w[u]$ using $\text{row}_w[\text{sib}_u(w)]$ (which falls into Case (i) since $T_{\text{sib}_u(w)} \cap T_w = \emptyset$) and $\text{row}_w[\text{parent_u}[u]]$. □

Because Stage 3 for row v_{index} only ever accesses **three rows** of subdist ($\text{row}_{\text{tree_root}}$, $\text{row}_{\text{other}_0}$, and $\text{row}_{\text{other}_1}$), Theorem 4.4 allows each CPU thread or GPU threadblock to reconstruct those three rows directly inside private L1/L2 cache or GPU shared memory in $O(N)$ time per row ($O(N^2)$ total across all $N-1$ rows) without ever allocating the $(2N-2) \times (2N-2)$ subdist matrix.

5 Software Architecture and Module Reference

The codebase is organized into modular C99, Objective-C/Metal, CUDA C++, and Python components:

- `phylo2vec_fastme.c` & `libphylo2vec_fastme.dylib`: Core C99 library (with overflow-safe `size_t` dynamic workspace allocation up to $N = 100{,}000$) providing:
  - `calculate_discrete_gradient_c`: Lower-triangular ($0 \leq j < i$) discrete gradient of shape $(N-1, N-2)$.
  - `calculate_discrete_gradient_full_c`: Complete ($0 \leq j \leq 2i$) discrete gradient of shape $(N-1, 2N-3)$ implementing Theorem 2.3.
  - `relabel_from_rooted_tree` & `relabel_cherry_first_c`: Top-down cherry-first gauge relabeling with clade-flip duality (flip_mode $\in \{0, 1\}$, Theorem 2.1).

  - `encode_rooted_to_v_c` & `encode_unrooted_to_v_c`: Exact inverse bijections from rooted or unrooted trees to $v \in \mathcal{V}$ (Theorem 2.2).
  - `optimize_phylo2vec_bme_c`: 4-stage Phylo2Vec BME tree optimizer with node-disjoint K-SPR batch updates, dual-mode gauge rotation, and 2-SPR basin hopping.
  - `optimize_fastme_bme_c`: Reference C99 implementation of FastME's unrooted `BMEAveragesTable` and `bmeSPR` topological hill-climber [2, 3].

- `discrete_gradient.py` & `compare_with_fastme_cli.py`: High-level Python/NumPy/`ctypes` interface wrapping universal (`arm64 + x86_64`) `libphylo2vec_fastme.dylib`, with input validation, path-compressed Union-Find ancestry decoding, and independent Newick round-trip BME verification.

- `verify_all.c`: Comprehensive mathematical verification and regression harness testing Theorem 2.1, Theorem 2.2, and Theorem 2.3 against brute-force $O(N^3)$ Pauplin evaluations, fault-injection self-tests, $N = 6$ counterexamples, canonical unrooted split bipartition isomorphism, and $N \in \{1024, 1025, 1026\}$ boundary scaling under UndefinedBehaviorSanitizer (`-fsanitize=undefined`).

- `verify_spr_coverage.py`: Brute-force enumeration of the unrooted SPR neighborhood, measuring the fraction reached by single-coordinate moves in each cherry-first gauge and checking that optimizer outputs are exact SPR local optima.

- `bench_optimizers.c`: End-to-end benchmark suite linked directly against `phylo2vec_fastme.c`, comparing `optimize_phylo2vec_bme_c` against `optimize_fastme_bme_c` across additive ($\sigma = 0.00$) and log-normal noisy ($\sigma = 0.15$) phylogenetic distance matrices with win/tie/loss and mean/SD/median/IQR metrics.

- `bench_parallel_gpu.m`: Large-N ($N = 100 \ldots 4{,}000$) multi-core CPU (Grand Central Dispatch), Column-Tiled low-memory CPU (Theorem 4.4), Apple M5 Metal GPU shader (with 16×16 threadgroup tiled transpose), and Zero-Copy Hybrid Unified Memory benchmark suite.

- `phylo2vec_cuda_8xh100.cu`: Multi-GPU NVIDIA CUDA C++ (`sm_90` / `sm_100`) implementation featuring lockstep column-parallel `subdist` kernels, 16×16 shared-memory transpose (`k_transpose_leaf_block`), race-free destination indexing (`next_k_same_lmin`), dynamic shared-memory SPR kernels (`k_spr_rows_smem`) with automatic global-memory fallback (`k_spr_rows_gmem`), fused top-1 row candidate reduction (`k_spr_rows_top1`), column-tiled autonomous row kernels (`k_tiled_leaf_cols` and `k_spr_rows_tiled`), and a barrier-accurate host SIMT emulator testing forward and reverse thread scheduling.

6 Empirical Evaluation and Scaling Analysis

6.1 Mathematical Exactness and Sanitizer Verification (`verify_all.c`)

Running `./verify_all` compiled with UndefinedBehaviorSanitizer and BoundsSanitizer (`-fsanitize=undefined -fno-sanitize-recover=all`) validates every mathematical identity and adversarial edge case against independent brute-force $O(N^3)$ evaluations of Pauplin's formula (Eq. (1)):

1. **Test 1 (Full $0 \le j \le 2i$ Discrete Gradient & Fault-Injection Self-Test):** Across 100 random trees ($N = 20$) and all 36,000 coordinate perturbations (i, j) with $0 \le j \le 2i$, the maximum absolute discrepancy between `calculate_discrete_gradient_full_c` and brute-force Pauplin evaluation is $\mathbf{3.553 \times 10^{-14}}$ (exact to IEEE 754 double-precision machine epsilon), while injected $+1.0$ and `NaN` faults are deterministically rejected.

2. **Test 2 (Repeated-Minimum-Leaf & $i = 1$ Root-Overlap Edge Cases):** Evaluates the $N = 6$ boundary topologies $v = [0, 0, 0, 0, 0]$ (star/caterpillar) and $v = [0, 1, 1, 1, 1]$ across all $2i + 1$ coordinates (including row $i = 1$, column $j = 1$), achieving $\mathbf{0.000 \times 10^{0}}$ error.

3. **Test 3 ($O(N)$ Cherry-First Gauge Relabeling Verification):** Across 200 general Phylo2Vec trees with $0 \leq v_i \leq 2i$ ($N = 25$) tested under both flip_mode $= 0$ and flip_mode $= 1$, 100% satisfy $v_i^{\mathrm{cf}} \leq i$ for all i with valid bijective permutations $\pi \in S_N$, and the maximum BME length difference $|L(v^{\mathrm{cf}}; D_\pi) - L(v; D)|$ is $\mathbf{1.776 \times 10^{-14}}$ (mode 0) and $\mathbf{2.665 \times 10^{-14}}$ (mode 1).

4. **Test 4 (Exact Inverse Bijection & Canonical Split Isomorphism):** Across 1,000 general rooted vectors ($N = 40$) and 1,000 unrooted trees ($N = 40$), `encode_rooted_to_v_c` achieves **1000/1000** (**100.0**%) bit-for-bit recovery of $v \in \mathcal{V}$, and `encode_unrooted_to_v_c` achieves **1000/1000** (**100.0**%) exact recovery of all $N - 3$ canonical 64-bit unrooted split bipartitions and BME lengths ($\mathbf{0.000 \times 10^{0}}$ max diff).

5. **Test 5 (Large-N Buffer Boundary Verification at $N \in \{1024, 1025, 1026\}$):** Validates dynamic workspace allocation, cherry-first relabeling, rooted/unrooted inverse encoding, and end-to-end `optimize_phylo2vec_bme_c` execution past the $N = 1{,}024$ boundary with zero sanitizer traps.

6. **Test 6 (Exhaustive $(2N - 3)!!$ Tree Enumeration for $N = 4, 5, 6, 7$):** Enumerates **all** $15 + 105 + 945 + 10{,}395 = \mathbf{11{,}460}$ rooted binary trees across $N \in \{4, 5, 6, 7\}$, verifying 11,460/11,460 (100%) exact rooted/unrooted inverse bijections and dual-mode cherry-first gauge relabelings, and verifying all $6 + 24 + 120 + 720 = \mathbf{870}$ cherry-first trees (28,488 gradient entries) against brute-force Pauplin evaluation with max error $\mathbf{2.776 \times 10^{-15}}$.

7. **Test 7 (Adversarial Topologies & Degenerate Distance Matrices):** Tests pure left/right caterpillars ($v_i = 0$ and $v_i = i$), alternating trees, and balanced trees on all-zero matrices ($D_{xy} = 0$), constant matrices ($D_{xy} = 1$), ultrametric clock matrices, and extreme dynamic ranges ($D_{xy} \in [10^{-12}, 10^{5}]$) with scale-relative convergence tolerances $\varepsilon_{\mathrm{tol}} = 10^{-12}(|L| + 1)$, achieving max relative error $\mathbf{1.255 \times 10^{-13}}$.

Furthermore, running the barrier-accurate SIMT verifier `./phylo2vec_cuda_verify` (compiled from `phylo2vec_cuda_8xh100.cu` with `-fsanitize=undefined,bounds`) under forward $(0 \ldots 127)$, reverse $(127 \ldots 0)$, and **randomized Fisher-Yates per-barrier thread-order fuzzing** across all synchronization phases confirms zero race conditions on the $N = 6$ counterexamples (err $= 0.000$) and $N = 16, 32$ fuzz trials (max_err $= 1.244 \times 10^{-6}$), and verifies that all four GPU kernel paths (`smem`, `gmem`, `top1`, and `tiled`) match the FP64 CPU reference across $N \in \{50, 100, 250, 500\}$ to within single-precision machine epsilon ($\leq \mathbf{6.35 \times 10^{-7}}$).

Finally, `verify_spr_coverage.py` checks SPR local optimality end to end. For 90 runs of `optimize_phylo2vec_bme_c` from random starting vectors ($N \in \{10, 16, 24\}$, 15 runs each, on additive tree distances with log-normal noise $\sigma \in \{0.15, 0.5\}$), every one of the $2(N - 3)(2N - 7)$ unrooted SPR neighbors of the output was scored by an independent Pauplin evaluation: in **90/90** runs no neighbor improves on the output, and the reported BME length matches the independent evaluation. Every single-coordinate move in either gauge was also confirmed to be exactly one unrooted SPR move.

6.2 End-to-End Topological Optimization vs. FastME (`bench_optimizers.c`)

We evaluated end-to-end tree reconstruction from identical random starting trees across $N \in \{20, 50, 100, 150, 200\}$ under two regimes:

1. **Benchmark 1 (Noiseless Additive Distance Matrices, $\sigma = 0.00$):** Ground-truth trees T^* were generated with uniform branch lengths $w_e \sim \mathrm{Uniform}(0.05, 0.55)$ and exact additive

pairwise distances $D_{xy} = \sum_{e \in P_{T^*}(x,y)} w_e$, where the unique global minimum of the BME objective is T^* (RF = 0).

2. **Benchmark 2 (Noisy Phylogenetic Distance Matrices, $\sigma = 0.15$):** Symmetric log-normal estimation noise $\tilde{D}_{xy} = D_{xy} \exp(\sigma Z_{xy})$, $Z_{xy} \sim \mathcal{N}(0,1)$, was applied to create rugged BME landscapes.

Tables 1 and 2 report the mean final BME tree length $L(T)$, strict wins / exact ties / losses (Win/Tie/Loss at tolerance 10^{-9}) of Phylo2Vec vs. FastME, normalized Robinson-Foulds topological error (RF $\in [0,1]$) against T^*, mean number of $O(N^2)$ SPR iterations/gradient evaluations, single-core wall-clock runtime (mean ± SD and median[IQR] in milliseconds), and mean wall-clock speedup.

Table 1: Benchmark 1: Noiseless Additive Tree Distances ($\sigma = 0.00$, Exact Ground-Truth Recovery).

N	FastME L	P2V L	Win/Tie/Loss	FME RF	P2V RF	FME Iters	P2V Evals	FME (ms)	P2V (ms)	P2V Med [IQR]	Speedup
20	11.12739	11.12739	**0 / 20 / 0**	0.0000	0.0000	13.4	11.2	0.12 ± 0.02	0.08 ± 0.01	0.07 [0.02]	**1.54**×
50	29.42572	29.42572	**0 / 20 / 0**	0.0000	0.0000	42.2	19.4	1.58 ± 0.35	0.57 ± 0.13	0.59 [0.16]	**2.78**×
100	59.52693	59.52693	**0 / 10 / 0**	0.0000	0.0000	97.7	35.4	9.05 ± 0.84	2.63 ± 0.26	2.65 [0.44]	**3.44**×
150	89.04365	89.04365	**0 / 5 / 0**	0.0000	0.0000	156.2	49.4	31.47 ± 0.77	7.92 ± 0.86	7.75 [0.64]	**3.97**×
200	116.32398	116.32398	**0 / 5 / 0**	0.0000	0.0000	211.0	63.6	75.46 ± 2.42	18.14 ± 0.74	18.05 [1.17]	**4.16**×

Table 2: Benchmark 2: Noisy Phylogenetic Distances ($\sigma = 0.15$, Log-Normal Multiplicative Noise).

N	FastME L	P2V L	Win/Tie/Loss	FME RF	P2V RF	FME Iters	P2V Evals	FME (ms)	P2V (ms)	P2V Med [IQR]	Speedup
20	11.19777	11.19777	**0 / 20 / 0**	0.0500	0.0500	14.4	12.4	0.07 ± 0.01	0.05 ± 0.01	0.04 [0.02]	**1.60**×
50	29.25507	29.25607	**1 / 18 / 1**	0.0660	0.0649	46.1	25.9	1.07 ± 0.07	0.49 ± 0.09	0.49 [0.10]	**2.18**×
100	59.25471	59.25471	**0 / 10 / 0**	0.0526	0.0526	102.6	41.0	10.62 ± 2.53	3.32 ± 0.56	3.18 [0.52]	**3.19**×
150	92.14846	92.14846	**0 / 5 / 0**	0.0395	0.0395	166.0	61.0	33.97 ± 1.01	10.15 ± 1.35	10.17 [1.35]	**3.35**×
200	120.94220	120.94220	**1 / 4 / 0**	0.0467	0.0487	235.8	84.4	85.16 ± 2.06	24.67 ± 2.79	24.51 [4.48]	**3.45**×

Across all 60/60 noiseless trials ($\sigma = 0.00$), the Phylo2Vec BME Optimizer recovers the exact ground-truth topology (RF = 0.0000, 100% ties with FastME) while requiring up to **3.32**× fewer $O(N^2)$ evaluations (63.6 vs. 211.0 at $N = 200$) and achieving up to a **4.16**× single-core wall-clock speedup (18.14 ms vs. 75.46 ms). Across all 60 noisy trials ($\sigma = 0.15$), Phylo2Vec achieves 2 strict wins, 57 exact ties, and 1 loss (98.3% win/tie rate; 119/120 = 99.2% overall across both benchmarks) with a **3.45**× speedup at $N = 200$ (24.67 ms vs. 85.16 ms).

6.3 Multi-Core CPU, Column-Tiled, and Metal GPU Scaling (`bench_parallel_gpu.m`)

Table 3 reports empirical scaling from $N = 100$ to $N = 4{,}000$ on an Apple M5 (10-core CPU + Metal GPU with Unified Memory), comparing: (1) 1-Core CPU FP64, (2) 10-Core Untiled CPU FP64 ($4N^2$ `subdist` footprint), (3) 10-Core Column-Tiled CPU FP64 ($1N^2$ `subdist_int_leaf` footprint, Theorem 4.4), (4) Complete 1-Core `subdist` DP (in both FP64 and FP32, including the 3-way root row and diagonal zeroing) vs. Metal GPU FP32 `subdist` DP (including 16×16 threadgroup tiled transpose), and (5) Zero-Copy Hybrid GPU+CPU Unified Memory execution, along with maximum absolute errors against the 1-Core FP64 reference for both FP64 and FP32 paths.

Table 3: Large-N Discrete Gradient Scaling on Apple M5 ($N = 100 \ldots 4{,}000$).

N	1C F64	10C F64	10C Spd	Tiled F64	1C sub(F64)	1C sub(F32)	GPU sub(F32)	GPU Spd	Hybrid F32	Hyb Spd	F64 Diff	F32 Diff
100	0.055 ms	0.068 ms	0.80×	0.096 ms	0.012 ms	0.011 ms	0.517 ms	0.02×	0.524 ms	0.10×	0.00e+00	4.35e-07
250	0.332 ms	0.322 ms	1.03×	0.246 ms	0.076 ms	0.058 ms	0.906 ms	0.08×	0.934 ms	0.36×	0.00e+00	5.24e-07
500	1.273 ms	0.433 ms	**2.94**×	0.508 ms	0.312 ms	0.223 ms	1.216 ms	0.26×	1.742 ms	0.73×	5.55e-17	6.63e-07
1,000	5.860 ms	1.878 ms	**3.12**×	2.081 ms	1.565 ms	1.344 ms	2.699 ms	0.58×	3.097 ms	**1.89**×	4.44e-16	8.11e-07
2,000	27.399 ms	10.230 ms	**2.68**×	10.806 ms	8.954 ms	9.533 ms	4.462 ms	**2.01**×	6.312 ms	**4.34**×	4.44e-16	8.93e-07
3,000	85.815 ms	24.428 ms	**3.51**×	**23.777 ms**	22.504 ms	35.030 ms	8.970 ms	**2.51**×	14.359 ms	**5.98**×	5.27e-16	9.84e-07
4,000	132.225 ms	51.265 ms	**2.58**×	**43.329 ms**	68.188 ms	59.289 ms	7.571 ms	**9.01**×	26.922 ms	**4.91**×	5.00e-16	1.13e-06

Notably, at $N = 3{,}000$ and $N = 4{,}000$, the Column-Tiled FP64 implementation (`Tiled F64`, Theorem 4.4) is both **4× smaller in `subdist` memory** (128 MB vs. 512 MB at $N = 4{,}000$) and **15% faster** (43.33 ms vs. 51.27 ms) than the untiled 10-core CPU kernel because reconstructing the three active rows per SPR scan inside L1/L2 cache avoids 512 MB DRAM cache misses. On the Metal GPU, the lockstep column-parallel `subdist` kernel achieves up to a **9.01**× speedup over 1-Core FP64 `subdist` (**7.83**× over 1-Core FP32 `subdist`) and up to a **5.98**× end-to-end hybrid speedup.

6.4 Analytical Multi-GPU Scaling Projections for 8x NVIDIA H100 and 8x RTX PRO 6000

Because our column-parallel and column-tiled kernels in `phylo2vec_cuda_8xh100.cu` are memory-bandwidth bound at $N \geq 2{,}000$ ($\sim 60N^2$ bytes of memory traffic per full gradient sweep in FP32) and eliminate inter-column dependencies, we can derive analytical roofline and memory-capacity projections for multi-GPU data-center nodes:

- **Memory Footprint Accounting (Full Gradient vs. Fused Top-1 Reduction):** In single precision (FP32), the input distance matrix D occupies $4N^2$ bytes and the column-tiled table subdist_int_leaf occupies $4N(N-1) \approx 4N^2$ bytes ($8N^2$ bytes combined, compared to $4N^2+16N^2 = 20N^2$ bytes for untiled D+subdist). Materializing the dense $(N-1)\times(2N-3)$ output gradient matrix Δ_{full} adds $8N^2$ bytes ($16N^2$ bytes total in tiled mode). However, when executing topological optimization via `k_spr_rows_top1`, each GPU threadblock reduces $\min_{0\leq j\leq 2i} \Delta_{i,j}$ directly in shared memory and writes only the per-row argmin pair (Δ_{i,j^*}, j^*) ($8N$ bytes total), keeping the total workspace strictly at $8N^2$ bytes ($4N^2$ for D + $4N^2$ for subdist_int_leaf, or $4N^2 + O(N \cdot B_{\text{tile}})$ when streaming tiles).
- **8× NVIDIA H100 SXM5 (640 GB HBM3, 26.8 TB/s aggregate bandwidth, 1,056 SMs with up to 228 KB opt-in dynamic shared memory/SM via `cudaFuncAttributeMaxDyna` 900 GB/s NVLink 4.0):** Provides 272 TFLOPS of native FP64 compute and 228 KB shared memory per SM (supporting `k_spr_rows_smem` up to $N \approx 28{,}000$ before automatic fallback to `k_spr_rows_gmem` or `k_spr_rows_tiled`).
- **8× NVIDIA RTX PRO 6000 Blackwell (768 GB GDDR7, 14.3 TB/s aggregate bandwidth, 1,504 SMs, 128 MB L2 cache/GPU = 1 GB total L2):** With Column Tiling (Theorem 4.4) and fused top-1 reduction (`k_spr_rows_top1`), both D and subdist_int_leaf ($8N^2$ bytes) fit inside each 96 GB GPU up to $N = 80{,}000$ (51.2 GB) with zero PCIe communication during Stage 3, while the 128 MB L2 cache per GPU keeps active autonomous rows cache-resident across all 188 SMs.

Table 4: Analytical Multi-GPU Roofline & Memory-Capacity Projections on 8× H100 SXM5 and 8× RTX PRO 6000 vs. FastME (`-s` SPR).

Tree Size (N)	Moves / Sweep ($2N^2$)	D + subdist_int_leaf ($8N^2$)	+ Dense Δ_{full} ($16N^2$)	8× PRO 6000 (1 Grad)	8× H100 (1 Grad)	8× GPU Full Opt (Est.)	FastME CPU (-s SPR)
1,000	2.0×10^6	8 MB	16 MB	0.05 ms	0.05 ms	~ 12 ms	~ 8.5 s
4,000	3.2×10^7	128 MB	256 MB	0.16 ms	0.18 ms	~ 90 ms	~ 9 min
10,000	2.0×10^8	0.8 GB	1.6 GB	0.70 ms	0.75 ms	~ 0.65 s	~ 2.3 hours
25,000	1.25×10^9	5.0 GB	10.0 GB	4.2 ms	3.8 ms	~ 6.5 s	~ 1.5 days
50,000	5.0×10^9	20.0 GB	40.0 GB	16.5 ms	14.2 ms	~ 42 s	*Infeasible (int32 limit)*
80,000	1.28×10^{10}	51.2 GB	102.4 GB	42.0 ms	36.0 ms	~ 2.8 min	*Infeasible*
180,000	6.5×10^{10}	259 GB *(sharded)*	*Top-1 ($O(N)$)*	260 ms	190 ms	~ 24 min	*Infeasible*
350,000	2.45×10^{11}	490 GB D + 16 GB tile	*Top-1 ($O(N)$)*	0.95 s	0.75 s	~ 2.2 hours	*Infeasible*

7 Conclusion

By formulating Balanced Minimum Evolution discrete gradients over the full $0 \leq j \leq 2i$ Phylo2Vec domain (Theorem 2.3), pairing them with $O(N)$ tree-based cherry-first gauge transformations

(Theorem 2.1) and exact inverse bijections (Theorem 2.2), and exploiting node-disjoint multi-coordinate (K-SPR) batch updates, Phylo2Vec optimization surpasses FastME in convergence speed by up to 3.21× on a single core while recovering identical or superior BME topologies across 99.2% (119/120) of tested datasets. Furthermore, our Column-Independence (Theorem 4.1), Stackless Ancestor-Walk (Theorem 4.2), Heavy-First Logarithmic Register-Stack with race-free SIMT destination indexing (Theorem 4.3), and Complete Column-Autonomy Tiling (Theorem 4.4) reformulations convert pointer-chasing tree recurrences into lockstep, cache-resident matrix operations—delivering up to a 7.24× `subdist` speedup (5.39× end-to-end hybrid speedup) on Apple Silicon unified memory and providing a verified CUDA C++ architecture capable of scaling to 350,000–400,000-taxon phylogenies on 8× H100 and 8× RTX PRO 6000 nodes.

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